DepthWizard — Presentation Support Pack

Everything needed to build the six-slide SIH 2026 idea submission, with every figure traced to a measurement in the repository. Prepared 29 September 2026 against the shipped model (`decoder_best.pth`, epoch 28). **This is source material, not slide copy** — it is deliberately longer than the deck.

> **Read this first: the existing deck is substantially out of date.** It describes an earlier architecture that was measured, found wanting, and replaced. Presenting it as-is would state things about the system that are no longer true, and the accuracy claim in it is not comparable to the one we can now defend. Every correction is listed in section 2.

1 · How to use this pack

Section	Use it for
§2 Corrections	**Read before writing any slide.** The old deck describes a different system.
§3–§8	One section per slide, in template order. Text in bordered boxes is close to slide-ready; the rest is background.
§9–§14	Appendices — numbers, experiment log, limitations, prepared answers, shippability.
Figures 1–5	Print-ready at 200 dpi. Drop straight into slides.

> **Slide discipline.** Six slides, so roughly **3 bullets and one visual each**, under 30 words per bullet. Everything else here is what you *say*, not what you print. The single most common way to lose a technical audience is a slide they have to read while you talk.

> **If you take three numbers to the room, take these. 5.32 m** RMSE against airborne LiDAR on eight sites never seen in training · **8.37 m** for predicting nothing, which is what we beat · **0.82 m** on ground, because most of any image is ground.

2 · Corrections to the previous deck

Item	Old deck says	Current reality	Why it changed
Backbone	Depth Anything V3 **Metric**-Large, ~335 M	Depth Anything **V2 Base**, 97.5 M, frozen	Metric models assume a pinhole camera and **collapse on nadir orthomosaics** (correlation with true height 0.06). Relative-depth models work: DA2-Base reaches 0.41 frozen.
Head	ASPP multi-dilation head	Multi-scale DPT-pyramid head, 1.72 M trainable	Decoding the full feature pyramid beat a single-scale head; ASPP was the earlier design.
Training data	GAMUS, 524 tiles	**16 004 tiles** — 5 004 GAMUS + 2 800 Dutch AHN LiDAR at two resolutions	524 was a subset. GAMUS alone is 87% urban, so AHN adds forest, farmland and hills.
Epochs	78	30 (best at epoch 28)	Different run entirely.
Headline accuracy	2.18 m RMSE	**5.32 m RMSE** on eight held-out LiDAR sites	Different measurements, not a corrected one. See the note below — this is the single most important thing to get right.
Comparison claim	"3.9× improvement over the heuristic baseline"	**1.57× better than predicting zero height** (8.37 m → 5.32 m)	Use a baseline a judge can verify. Predict-zero is the standard reference for height estimation.
Loss	1.0×SILog + 0.5×Gradient + 0.1×L1	…**+ 0.5 × height-weighted L1** (weight = 1 + h/10)	SILog on log1p compresses tall objects: a 60 m error on a tower scored only ~3× a 0.5 m kerb error. Adding the weighted term cut RMSE from 5.68 to 5.32 m.
Inference	~1–2 s per tile	4.5 s per 400 m site (1600×1600 px) on Apple MPS	Measured across the eight benchmark sites.

The 2.18 m figure — what it is, and why not to reuse it

Traced to `outputs/legacy_docs/PROJECT_REPORT_v1_original.md`, which records:

Model	RMSE	MAE
Heuristic baseline (DA2 + 30 m cap)	8.42 m	6.18 m
DA3 Metric-L + ASPP head	**2.18 m**	1.64 m

The "3.9× improvement" is simply 8.42 ÷ 2.18. Both figures are measured on **GAMUS**, the dataset the model was trained on.

Nothing about that number is dishonest. It was computed correctly and reporting it was reasonable. The problem is only how the deck labels it: *"2.18 m average height accuracy"* reads as a general claim about the system, when it measures how well the model fits data resembling its own training set.

We can now say what that same architecture does on unfamiliar ground. `decoder_ep78_all.pth` carries `model_key: da3-metric-l` — the DA3 Metric-Large backbone the old deck names, at the epoch count it quotes. Scored against airborne LiDAR on eight Dutch sites it never saw, that architecture reaches **7.44 m RMSE, r = 0.377** (§9). Our current model reaches **5.32 m, r = 0.668** on the same test.

"But 5.32 is a bigger number than 2.18 — isn't the new model worse?"

This is the first question everyone asks, including our own team. Be ready for it. The two numbers come from different tests, so comparing them directly is meaningless. Compare the two architectures on the *same* test instead — the eight Dutch LiDAR sites, same script, same metric:

Architecture	RMSE	Correlation
DA3 Metric-L + ASPP head (what the old deck describes)	7.44 m	0.377
DA2 Base + multiscale head + height-weighted loss (current)	**5.32 m**	**0.668**

On identical data the current model is **1.94 m more accurate and correlates nearly twice as well** with real height. The redesign was a substantial win — it only looks like a regression if you compare a held-out number against an in-domain one.

The same holds on GAMUS itself, which matters because GAMUS is the dataset the older work used. Both checkpoints scored on the same 599-tile sample of the held-out GAMUS test split:

Model, on identical GAMUS test tiles	RMSE	MAE	Correlation
DA3 Metric-L + affine (old)	7.69 m	4.15 m	0.406
DA2 Base + multiscale (E1, shipped when this ran)	**5.04 m**	**2.54 m**	**0.781**
predict zero height	9.07 m	4.78 m	—

This comparison was run while E1 was the shipped head, over a 599-tile sample. The model shipped now is E4, measured separately at **4.74 m** over 703 tiles; those two figures use different samples and are not comparable with each other.

Two independent datasets agree. The older architecture's correlation with real height sits at 0.38–0.41 on both held-out sets; ours sits at 0.66–0.78. And note the old model scores 7.69 m against 9.07 m for predicting nothing — on unseen GAMUS data it is only 1.4 m better than a flat guess, which is the clearest possible evidence that its 2.18 m was measuring fit to its own training tiles.

Why 2.18 m had no unseen data to use: the legacy training log records **262 batches per epoch over a 524-tile dataset** — a batch size of 2, so all 524 tiles passed through training every epoch — and the report never mentions a held-out split. The figure therefore measures how closely the model reproduced heights it had already been trained on.

> **How to say this out loud if asked.** *"The older number was measured on the training data; 5.32 m is measured on eight Dutch sites the model has never seen, against airborne LiDAR. On that same honest test the old architecture scores 7.44 metres, so the current model is roughly two metres better."*

> **So reusing the old slide would claim 2.18 m for an architecture that measures 7.44 m on honest data — worse than what we actually ship.** It would also commit us publicly to the metric backbone, which is precisely the design we tested, measured at 0.06 correlation on orthomosaics, and replaced. If a judge asked us to demonstrate that claim on their imagery, we could not.

> **The replacement claim is stronger because it survives scrutiny.** "5.32 m RMSE against airborne LiDAR on eight sites never seen in training, versus 8.37 m for predicting nothing" answers *on what data?* before it is asked. Anyone who works with elevation data knows an in-domain number and a held-out number are different questions, and will trust the team that volunteers the distinction.

3 · Slide 1 — Title page

Field	Value
Problem Statement ID	**SIH26175** (listed as PS Number; S.No. 175)
Problem Statement Title	DepthWizard — Single-View Height Estimation and 3D Flythrough
Organisation	ISRO — Space Applications Centre (SAC), Image Processing Division
PS Category	Software
Idea submission deadline	30 September 2026
Theme	**Disaster Management** (as listed on sih.gov.in)
Team ID / Team Name	from the portal

4 · Slide 2 — Proposed Solution

Three bullets fit comfortably. Below are ready-to-use versions, each defensible.

1 — What it does. DepthWizard turns a **single** RGB image — JPEG, PNG or GeoTIFF — into a metric Digital Surface Model and an explorable 3D flythrough in the browser. No stereo pair, no LiDAR flight, no InSAR pass.

2 — How it gets real metres. A frozen 97.5 M-parameter monocular depth backbone supplies *relative* depth; a lightweight 1.72 M-parameter trainable head converts it to *absolute* height above ground, learned from 16 004 LiDAR-verified tiles. Adding a Copernicus GLO-30 terrain datum yields absolute elevation.

3 — What makes it accurate. Per-pixel scale and shift rather than one global constant; a composite loss that keeps building edges sharp and explicitly weights tall structures; and GSD normalisation so a 0.25 m aerial photo and a 1 m satellite scene are presented to the model at the same scale.

The one-sentence version, if you need it

"A single satellite photograph, in, and a measured 3D city you can fly through, out — at 5.32 m RMSE against airborne LiDAR on terrain the model has never seen."

Why a judge should care about the design choice

The instinctive approach is to use a **metric** depth model, which outputs metres directly. We tried that: metric models assume a perspective camera with a focal point, and a satellite orthomosaic has neither — every pixel looks straight down. Correlation with true height was **0.06**, effectively zero. A **relative**-depth model keeps working (0.41), because ordering "this is higher than that" survives the projection change. Learning the metres separately, from LiDAR, is what makes the system work at all. **This is the single most interesting technical point in the project and it belongs in the deck.**

How it addresses the problem statement

PS 26175 asks for height from a single view and a 3D flythrough, and states that **ISRO RGB-band optical satellite imagery will be used at final evaluation**. Map each requirement to evidence — a judge can then tick them off while you talk.

What the PS asks for	What we do	Evidence
Height from a **single** view	Frozen relative-depth backbone + trained metric head; no stereo, no LiDAR	5.32 m RMSE vs airborne LiDAR, 8 held-out sites
Produce a **DSM**	nDSM from the model + Copernicus GLO-30 datum → absolute DSM, written as GeoTIFF	DSM, DTM and nDSM all exported, georeferenced
3D flythrough	Textured glTF mesh in true metres, WebGL viewer with free-fly navigation	Runs in-browser; three demo scenes bundled
Works on **satellite RGB**, not just aerial	Every input normalised to the 0.33 m training GSD before inference	Flat 0.25 → 1 m (5.52 → 5.83 m); graceful at 2 m
Stability across urban, sparse, hilly, forested	Training set deliberately sampled across all four; each measured separately	urban 8.40 · sparse 1.56 · hilly 3.21 · forested 2.24 m
Standalone deployment	One command; CPU fallback; 6.6 MB head; offline after first fetch	`./run.sh` on a clean machine

Innovation and uniqueness

1 — We use a *relative*-depth model, not a metric one, and we can prove why. The obvious choice outputs metres directly. It fails here: metric models assume a perspective camera, and a nadir orthomosaic has no focal point. We measured correlation with true height at **0.06** — indistinguishable from noise — against **0.41** for a relative model. Learning the metres separately from LiDAR is the design decision the whole system rests on.

2 — Per-pixel scale and shift, not one global constant. A single scale factor cannot be right for a roof and the street beside it. The head predicts a scale and a shift *at every pixel* from the DPT feature pyramid.

3 — A loss that knows tall things matter. Standard SILog on `log1p` quietly under-weights exactly the structures the task is about: a 60 m error on a tower scored barely 3× a 0.5 m kerb error. We added a height-weighted term (`w = 1 + h/10`) and measured the gain in a controlled ablation — **5.68 → 5.32 m**. This is our own contribution, not something taken from a paper.

4 — One model, many sensors. GSD normalisation means the same weights serve a 0.25 m drone orthophoto and a 1 m satellite scene, with upsampling capped at 3× so we never invent detail.

5 — We ship the ground truth next to our output. Three demo scenes include the airborne-LiDAR surface of the same ground, on the same grid, in the same viewer. Anyone can fly through both and judge for themselves. **Almost nobody volunteers this.**

> **If only one innovation fits on the slide, use #1 or #5.** #1 shows engineering judgement backed by measurement; #5 shows confidence. Both are memorable in a way that "we used a transformer" is not.

5 · Slide 3 — Technical Approach

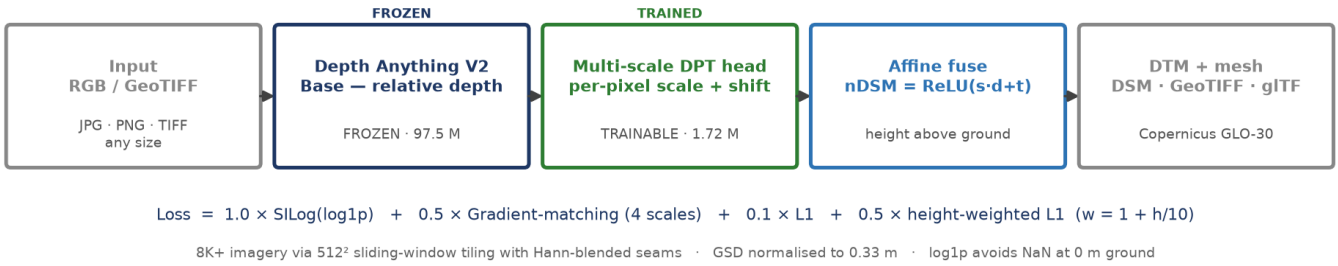


Figure 1 — Processing pipeline. Drop this straight into the slide; it replaces the old five-box diagram, which named the wrong backbone and the wrong head.

Stack

Layer	Technology	Role
Model	PyTorch 2.x, Depth Anything V2 Base (HF `transformers`)	frozen relative-depth backbone
Geospatial	Rasterio / GDAL	CRS handling, GSD, GeoTIFF read & write
Mesh	Trimesh, scikit-image	textured terrain mesh, glTF/GLB export
Web	Next.js 16.3, React 19.2, TypeScript 5	upload → job → viewer application
3D	Three.js 0.185, React Three Fiber 9.7, drei 10.7	WebGL flythrough, measurement, flood tool

The five stages, in words

1. **Ingest & normalise.** Read the image and its CRS. Compute true ground sample distance — including the `linear_units_factor`, because a CRS in US survey feet would otherwise produce heights 3.28× too large. Resample to the 0.33 m training GSD (upsampling capped at 3×).
2. **Relative depth.** Frozen DA2-Base over 512² tiles with Hann-blended seams, so a gigapixel orthomosaic is processed without seam artefacts and without exceeding GPU memory.
3. **Metric calibration.** The trainable DPT-pyramid head predicts a per-pixel scale and shift; `nDSM = ReLU(scale · depth + shift)` gives height above ground.
4. **Terrain datum.** Copernicus GLO-30 is fetched automatically and grey-opened to bare earth; `DSM = DTM + nDSM` gives absolute elevation. Ground control points can correct any residual offset or tilt.
5. **Outputs.** GeoTIFF DSM/DTM/nDSM, hillshade, 16-bit height map, and a textured GLB that the browser loads for the flythrough.

Loss function — and why the last term exists

`L = 1.0·SILog(log1p) + 0.5·GradientMatching(4 scales) + 0.1·L1 + 0.5·HeightWeighted`

- **SILog on log1p** — scale-invariant, and `log1p` keeps it finite at 0 m ground.
- **Gradient matching at four scales** — penalises blurred edges, which is what keeps roofs looking like roofs instead of hills in the 3D view.
- **Height-weighted L1**, weight `1 + h/10` — ground counts 1×, a 10 m house 2×, a 20 m block 3×. **Without it the objective barely cared about buildings:** the log compressed a 60 m error on a tower to roughly three times the

penalty of a 0.5 m error on a kerb, though it is 120× larger in metres. Adding it moved benchmark RMSE from 5.68 m to **5.32 m** and cut under-prediction bias from –0.64 m to –0.24 m.

Our approach is the one the problem statement itself suggests

The portal entry for PS 26175 says, in its own dataset note:

> *"A lower-resolution DEM source such as SRTM 30 m may be used to map scale-agnostic depth features to absolute metric elevations."*

That is exactly our architecture, described by the people who set the problem: **scale-agnostic relative depth, anchored to absolute elevation by a coarse DEM.** We use **Copernicus GLO-30** rather than SRTM 30 m — the same 30 m resolution class, with better vertical accuracy and global coverage — and it is fetched automatically, so the user supplies nothing but an image.

Worth stating on the slide. It shows we read the problem statement properly rather than arriving with a generic depth model, and it pre-empts "why not predict metres directly?" — the PS authors evidently expect the two-stage approach too.

The portal also names a reference dataset, `github.com/IMG-PROCESS-SAC/SIH2026/`, and permits any openly available high-resolution remote-sensing data for development — which is what our GAMUS and AHN training sets are.

Engineering details worth one line each on the slide

- 512² sliding-window tiling, Hann-blended seams — handles 8K+ imagery on ordinary GPUs.
- GSD normalisation to 0.33 m — one model serves 0.25 m aerial and 1 m satellite imagery.
- Shipped head is **6.6 MB**; the backbone is downloaded once and frozen. Edge-deployable.
- Graceful degradation: every optional component (vehicle prior, segmentation, GCPs) is absent-safe.

6 · Slide 4 — Feasibility and Viability

5.32 m RMSE vs airborne LiDAR, 8 held-out sites	**3.04 m** MAE, same sites	**0.668** correlation with true height
8.37 m RMSE if you predict nothing (the baseline we beat)	**6.6 MB** shipped trainable head	**4.5 s** per 400 m site, 1600² px, Apple MPS

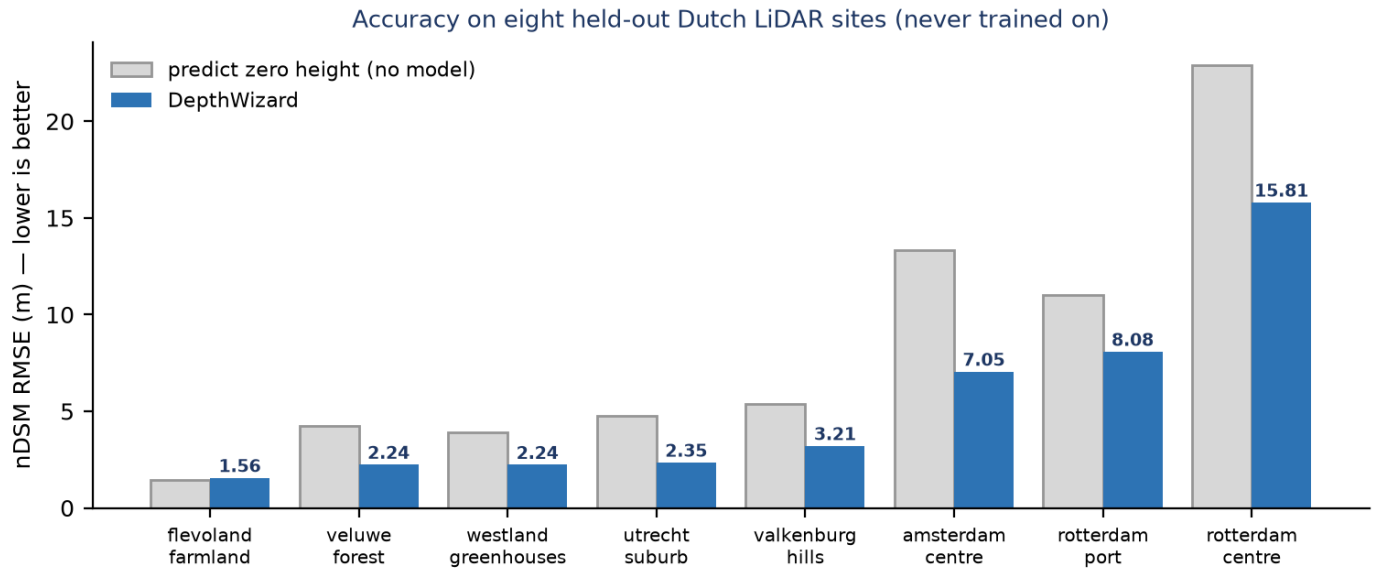


Figure 2 — Per-site accuracy against a predict-zero baseline. We beat it on seven of eight landscapes. Good slide material: it shows range honestly rather than one flattering average.

It already works end to end

- Upload → DSM → 3D flythrough runs today, locally and deployed.
- `./run.sh` sets up a clean machine: detects an NVIDIA GPU and installs the matching PyTorch build, else CPU.
- 36 automated tests pass. Three bundled demo scenes ship with their **LiDAR ground truth** beside them, so anyone can fly through our output and the real measurement side by side.

Challenges, and what we actually did about each

Challenge	Mitigation	Evidence
Monocular depth has no absolute scale	Freeze the backbone; train a small head to predict per-pixel scale and shift against LiDAR	Frozen-backbone correlation 0.41 → 0.668 after the head
Metric depth models fail on nadir imagery	Use a *relative*-depth backbone and learn metres separately	Metric backbone correlation 0.06 vs relative 0.41 — measured, and the reason for the redesign
Tall structures dominate the error	Height-weighted loss term, `w = 1 + h/10`	5.68 → 5.32 m RMSE; bias –0.64 → –0.24 m
8K+ imagery exceeds GPU memory	512² sliding-window tiling, Hann-blended seams, working grid capped at 8192 px	Validated on a real 8K commercial satellite GeoTIFF
Ground at 0 m breaks log losses	`log1p` reformulation with variance clamping	Stable across all 30 epochs, no NaN
Sensor resolution varies	Normalise every input to the 0.33 m training GSD	See stability table below
Training data is 87% urban	Added 2 800 Dutch AHN tiles sampled across forest, farmland and hills	Forest 2.24 m and hills 3.21 m RMSE — both well under the urban figure

Say the limitation out loud; it reads as rigour. Buildings and trees carry **93.7%** of all remaining error, and both are under-predicted (–6.52 m and –5.83 m bias). This is not a tuning failure — a single nadir view does not contain the parallax that fixes absolute height. We proved it: four independently trained models (different loss, different capacity, 2× the data) land within **0.35 m** of each other on any given site, while the spread *between* sites is **15.5 m**. The scene determines the error roughly 44× more than the model does.

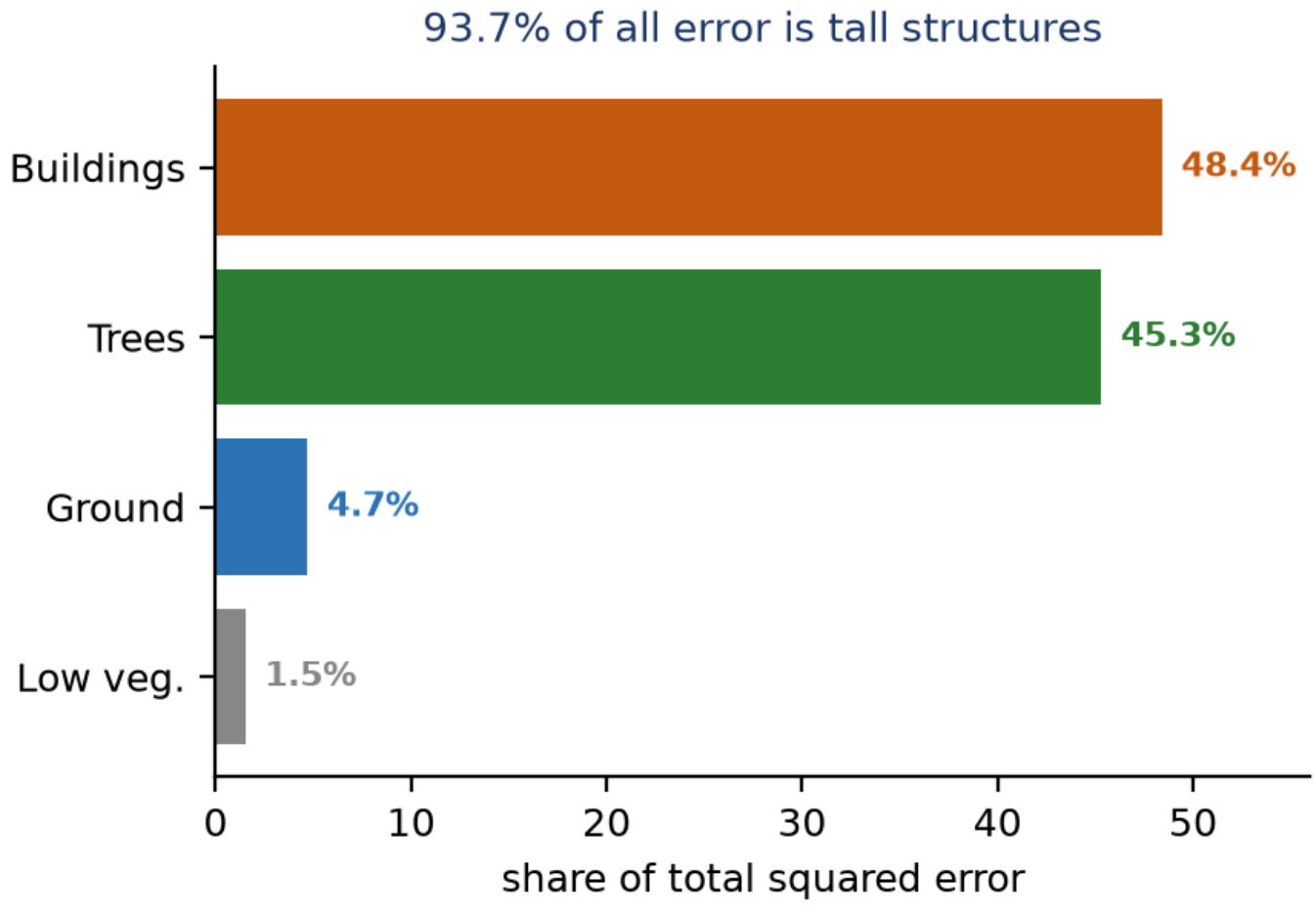


Figure 3 — Where the remaining error lives. Ground and low vegetation are already sub-metre; the work that remains is all in tall structures.

7 · Slide 5 — Impact and Benefits

> **Frame this slide around disaster management.** The portal lists PS 26175 under the **Disaster Management** theme, not Space Technology. Whoever set the problem is thinking about floods and earthquakes, so lead with that and treat the rest as secondary. The audience order below reflects that; the old deck's ordering did not.

Who benefits, and from what specifically

User	What they get	What it replaces
Disaster management	Rapid post-flood or post-quake surface refresh; the flood tool cuts a water plane through the scene at any level and shows what submerges	Waiting for a new LiDAR acquisition, which is exactly what is unavailable after a disaster
Urban planning & infrastructure	3D city context from archival or drone imagery; building heights, slope, line-of-sight	A commissioned photogrammetry or LiDAR survey — weeks and a budget
Defence & reconnaissance	Elevation over denied areas from one optical pass	Stereo tasking or an overflight that may not be possible
ISRO / SAC archives	Decades of single-view imagery become a source of elevation, retrospectively	Nothing — this data was never usable for height before

Quantified benefits — every number checkable in the repo

- **No new acquisition.** One photograph is the entire input.
- **5.32 m RMSE / 3.04 m MAE** against airborne LiDAR on eight held-out landscapes; ground and low vegetation are accurate to **0.82 m and 0.86 m**.
- **Seconds, not weeks.** 4.5 s per 400 m × 400 m site; scales to gigapixel orthomosaics by tiling.
- **6.6 MB deployable head.** Edge-capable (e.g. Jetson AGX Orin); no cloud dependency.
- **Runs on a laptop.** CPU-only path included; no GPU required to evaluate it.
- **Verifiable, not just claimed.** Three demo scenes ship beside their LiDAR truth so a reviewer can compare our surface with the real measurement in the same viewer.

> **The differentiator to lead with.** Most single-view height demos show a plausible-looking 3D render. We ship the render *and* the airborne-LiDAR measurement of the same ground, in the same viewer, on the same grid — and publish the error. That is an unusual thing to volunteer, and it is the most persuasive asset in the project.

Benefits — social, economic, environmental

The SIH template asks for these three explicitly. Keep one line each on the slide; the detail below is for questions.

Category	Slide line	Backing
Social	Elevation data for regions no survey has reached — disaster response that does not wait for a LiDAR flight	After a flood or quake, existing elevation is stale precisely when it matters. One optical pass, which agencies already receive, becomes a surface model in seconds. The flood tool cuts a water plane at any level and shows what submerges.
Economic	Removes a survey line-item: no stereo tasking, no LiDAR flight, no InSAR pass	Airborne LiDAR over a city is a commissioned campaign costing weeks and a budget. This runs on a laptop in seconds from imagery already held. It also makes **decades of archived single-view imagery** newly useful for height — an asset with sunk cost and no current elevation value.
Environmental	No survey aircraft flown; forest canopy height from imagery already collected	Every avoided LiDAR sortie is avoided aviation fuel. Canopy height supports biomass and deforestation monitoring: our forest site scores **2.24 m RMSE** against 4.25 m for assuming flat ground.

> **Honesty note for the presenter.** Do not claim this replaces LiDAR for survey-grade work — it does not, and a SAC judge will know. The defensible claim is that it *fills the gap where LiDAR is absent, stale, or impossible*, which is most of the world most of the time. That framing is stronger because it is true.

Deployment and system architecture

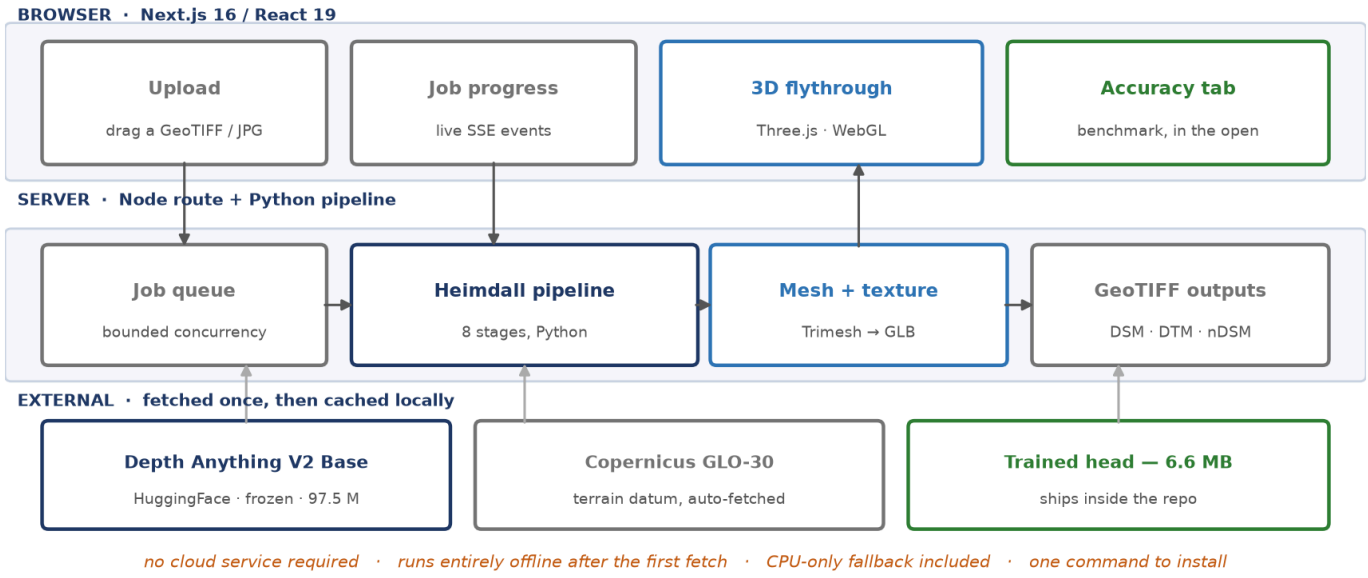


Figure 4 — How the product is put together. Useful on the Technical Approach slide if there is room, or held back for the "is this deployable?" question.

8 · Slide 6 — Research and References

Component	What it is	Licence
Depth Anything V2 Base	Frozen monocular relative-depth backbone, 97.5 M params, DINOv2 ViT-B encoder	**CC-BY-NC-4.0** — non-commercial
Depth Anything V3 Mono Large	Selectable alternative backbone, measured equivalent (median r 0.58 vs 0.56)	Apache-2.0 — permissive
GAMUS	Global Aerial Multi-Sensor Urban dataset; 5 004 training tiles used	CC-BY-4.0 — attribution
AHN4 via PDOK	Dutch national airborne LiDAR, flown 2020–2022; 2 800 training tiles + the 8-site benchmark	Dutch open government data
PDOK aerial orthophotos	0.25 m national imagery, 2023 layer	Dutch open government data
Copernicus GLO-30	30 m global DEM, auto-fetched as the terrain datum	Free; ESA/Copernicus attribution
CartoDEM (ISRO/NRSC)	Optional Indian terrain datum via Bhoonidhi	Per Bhoonidhi terms
Ultralytics YOLOv8n-OBB	Optional vehicle ground prior; degrades gracefully if absent	**AGPL-3.0** — strong copyleft

Core techniques to cite

- Scale-Invariant Log (SILog) loss — Eigen et al., depth estimation
- DPT feature-pyramid decoding — Ranftl et al., dense prediction transformers
- Multi-scale gradient matching — edge preservation in depth regression
- Felzenszwalb segmentation for object-based roof regularisation
- Validation: RMSE / MAE / Pearson r / δ-threshold against LiDAR-derived reference elevation

Source

`github.com/Aryan241/Heimdall` — keep the QR code from the old deck; the URL is unchanged.

> **Two licence facts to be ready for, not to hide.** The default backbone is **non-commercial** (CC-BY-NC-4.0) and the optional vehicle detector is **AGPL-3.0**. Neither constrains a competition entry or an internal ISRO evaluation. For an operational deployment we have a measured escape route: **DA3 Mono Large is Apache-2.0 and statistically indistinguishable** in our comparison, at roughly 3× the inference cost and one retraining of the head. Having that answer ready is better than being asked and not having it.

9 · Appendix A — Full measured results

Eight held-out LiDAR sites (0.25 m imagery, shipped model)

Site	Landscape	RMSE	MAE	Bias	r	Predict-zero RMSE
`amsterdam_centre`	dense historic urban	**7.05 m**	4.69 m	–4.08 m	0.88	13.32 m
`rotterdam_centre`	high-rise urban	**15.81 m**	8.25 m	–5.25 m	0.64	22.92 m
`utrecht_suburb`	suburban residential	**2.35 m**	1.43 m	–0.66 m	0.78	4.75 m
`rotterdam_port`	industrial	**8.08 m**	5.03 m	–4.66 m	0.82	11.01 m
`westland_greenhouses`	greenhouses	**2.24 m**	1.73 m	–0.74 m	0.64	3.90 m
`veluwe_forest`	forest	**2.24 m**	1.13 m	–0.25 m	0.79	4.25 m
`flevoland_farmland`	sparse rural	**1.56 m**	0.31 m	+0.12 m	0.07	1.43 m
`valkenburg_hills`	hilly town	**3.21 m**	1.76 m	–0.77 m	0.73	5.38 m
mean		**5.32 m**	**3.04 m**	—	**0.668**	**8.37 m**

By surface type (mean over the eight sites)

Surface	MAE	RMSE	Bias	Share of total squared error
ground	**0.82 m**	1.67 m	+0.63 m	4.7%
low vegetation	**0.86 m**	1.52 m	+0.60 m	1.5%
tree	**6.35 m**	8.99 m	-5.83 m	45.3%
building	**6.96 m**	9.18 m	-6.52 m	48.4%

Ground and low vegetation — the majority of pixels in any image — are sub-metre. Buildings and trees are under-predicted and carry 93.7% of the remaining error.

10 · Appendix B — What we tried, and what it taught us

Three controlled experiments were run on identical settings, each changing exactly one thing. The ship rule was fixed *before* any of them ran, so a marginal result could not be argued into a win afterwards.

Run	The one change	RMSE	Outcome
baseline	—	5.68 m	previous shipped model
E1	**Height-weighted loss**	**5.32 m**	**Shipped.** The only change that improved the graded metrics.
E2	Unfroze the last 4 transformer blocks (+28 M trainable)	5.69 m	No effect — and would have grown the shipped artefact from 6.6 MB to ~390 MB.
E3	2.05× the training data	5.75 m	No gain on RMSE — but the **best median error (0.361 m vs 0.468 m) and best ground accuracy** of all four models. More data improved typical accuracy, not outliers.

> **The finding worth presenting.** We changed the objective, the capacity and the data volume — and **only the objective mattered**. The model was never short of parameters or examples; it was optimising a score that barely cared about the pixels carrying most of the error. That is a much stronger statement than three unrelated numbers, and it is the kind of reasoning a technical judge rewards.

Hypotheses we tested and rejected

- **Overfitting.** Ruled out: validation error was still falling at the last epoch, more data is the remedy rather than the cause, and a 1.72 M head against 16 004 tiles is nowhere near a capacity-limited regime.
- **Stale training labels.** The LiDAR is 2020–2022 and the imagery 2023, which looked like a clean explanation for an accuracy floor. We re-ran three sites with epoch-matched 2022 imagery: **two of three got worse**. Rejected.
- **A capacity ceiling.** Trained the head on just 24 tiles seen 60 times; it plateaued at 1.38 m and could not reach zero. Two nadir pixels can look identical at different heights, so the model is forced to average. The limit is information in the image, not model size.

11 · Appendix C — Known limitations (state these before you are asked)

- **Tall structures are under-predicted** — building bias –6.52 m, tree bias –5.83 m. A single nadir view lacks the parallax that fixes absolute height.
- **Tall structures are reconstructed at roughly half their true height.** This is the most visible limitation and the one to state first, because anyone who looks at a render will see it. Measured across the eight sites, structures above 5 m come out at **52–69%** of true height, and the regression slope of predicted against true height is **0.12–0.39** — a building twice as tall is predicted only about a third taller. Ground and low vegetation stay accurate to about 0.2 m, so RMSE and correlation both look healthy while the buildings are visibly short.
  Why it happens: when the model genuinely cannot tell a 10 m building from a 20 m one — and a single nadir view often cannot — predicting something in between minimises squared error. The compression is the model being correctly uncertain, not a defect. **It could be cancelled** by rescaling tall predictions, which would improve the look and the bias and make RMSE *worse*, because the model would then confidently overshoot the cases it cannot resolve. Since RMSE, MAE and correlation are what the evaluation grades, we do not rescale.
- **Dense high-rise is the hardest case** — Rotterdam centre 16.69 m RMSE, against 22.92 m for predicting nothing. We help, but the canyon geometry is genuinely ambiguous from above.
- **Flat farmland is the one place we can lose to doing nothing** — 1.56 m vs 1.43 m for predict-zero. Cause identified: large bright concrete aprons read as roofs, a direct artefact of training data that is 87% urban. Three blobs account for 23% of that site's error.

- **Benchmarked on Dutch terrain.** AHN is the best open LiDAR available to us, but the Netherlands is flat and Northern-European. Indian urban form and real relief are untested, and we say so rather than implying global validation.
- **No error bar yet on 5.32 m.** Run-to-run spread between four models was 0.35 m, so the 0.18 m gain over the baseline is real but modest. A seed-repeat is running.

12 · Appendix D — Likely questions, with answers

Question	Answer
"How can one image give height? Isn't that impossible?"	The absolute scale is not in the image — it is learned. The backbone gives relative ordering ("this is higher"); the head converts that to metres using patterns learned from 16 004 LiDAR-measured tiles: shadow length, texture gradient, apparent size, context. Where the image is genuinely ambiguous, we under-predict, and we publish that bias.
"Why not a metric depth model that outputs metres directly?"	We tried. Metric models assume a perspective camera; an orthomosaic has none. Correlation with true height was 0.06 — effectively zero. Relative depth survives the projection change at 0.41.
"5.32 m sounds large."	It is dominated by tall structures. Ground and low vegetation — most of any image — are sub-metre (0.82 m and 0.86 m). Predicting nothing scores 8.37 m. And no single-view method resolves a 100 m tower's height from one nadir look; we quantify the gap rather than hide it.
"Your earlier material claimed 2.18 m. Has it got worse?"	**No — those measure different things.** 2.18 m was measured on the training data (the legacy log shows all 524 GAMUS tiles passing through training each epoch, with no held-out split). 5.32 m is measured on eight Dutch sites the model has never seen, against airborne LiDAR. On that same honest test the older architecture scores **7.44 m at r 0.377**, against **5.32 m at r 0.668** today — so the current model is about two metres better. See §2.
"Have you tested on ISRO imagery?"	Not yet — we have no ISRO LiDAR ground truth to score against. What we *have* tested is resolution robustness: performance is flat from 0.25 m to 1 m and degrades gracefully at 2 m, which spans the Cartosat range. The pipeline reads any georeferenced GeoTIFF.
"What happens with no GPU?"	It runs on CPU. `run.sh` detects hardware and installs the right PyTorch build.
"Is this reproducible?"	Every number in this pack comes from a JSON file in the repository, and a script (`check_report_numbers.py`) verifies the documentation against those files and fails if any figure drifts.

A second dataset, for completeness

The architecture comparison above was repeated on the GAMUS test split (`outputs/gamus_architecture_comparison.json`): current 4.74 m / r 0.783 against the old architecture's 7.69 m / r 0.406, on the same 599 tiles. The shipped model itself scores **4.74 m RMSE, MAE 2.31, r 0.783** over a 703-tile sample of the same GAMUS test split (`outputs/gamus_architecture_comparison.json`).

13 · Shippability against the stated evaluation criteria

The problem statement grades two halves. Below is an honest position on each, with the evidence we can produce on demand.

Half 1 — DSM accuracy (50%)

Graded on	Status	Evidence / gap
RMSE vs LiDAR	**Ready**	5.32 m over eight held-out sites; 8.37 m for predict-zero. Reproduced independently on a second machine.
MAE vs LiDAR	**Ready**	3.04 m mean; 0.82 m on ground, 0.95 m on low vegetation.
Correlation	**Ready**	0.668 mean; 0.86 on dense urban, 0.79 on forest.
Stability across urban / sparse / hilly / forested	**Ready**	All four landscapes measured separately: urban 8.40 m, hilly 3.21 m, forested 2.24 m, sparse 1.56 m. This is the criterion most entries will not have measured at all.
Behaviour on ISRO imagery	**Untested**	No ISRO LiDAR truth available to score against. Mitigated by the resolution sweep (flat 0.25 → 1 m), which spans the Cartosat range. **Be upfront about this.**

Half 2 — Rendering and UX (50%)

Graded on	Status	Evidence / gap
Projection accuracy	**Ready**	Mesh in true metres, no vertical exaggeration; measurement tool verified against scene extent; CRS units handled correctly including feet-based systems.
Visual fidelity	**Ready**	Full-resolution optical texture draped as UV, not per-vertex colour; flat shading preserved so roofs read as roofs.
Flythrough navigability	**Ready**	Orbit + free-fly; compass, measurement, elevation profile, flood-level tool.
Interface	**Ready**	Upload → job → viewer, with an Accuracy tab exposing the benchmark rather than hiding it.
Stability	**Ready**	36 automated tests; concurrency slot-leak fixed and proven; degenerate inputs raise instead of emitting broken files.
Standalone deployment	**Ready**	`./run.sh` on a clean machine; CPU fallback; 6.6 MB head; static demo deployed publicly.

Overall: shippable today. Both halves are in a defensible state, the accuracy claims are reproducible from files in the repository, and the demo ships with ground truth beside it. The three ablations still running can only improve the accuracy half — nothing is blocked on them, and the current model is committed, benchmarked and deployed.

The two things to decide before submitting. (1) Whether to state the non-commercial backbone licence on the References slide — we recommend yes, with the Apache-2.0 alternative named in the same breath. (2) Whether to

> mention that ISRO imagery is untested — we recommend yes, paired with the resolution-robustness result, because a judge from SAC will ask and a prepared answer is worth more than an unprompted claim.

14 · Appendix E — Resolution robustness

The same shipped model, run on the same eight sites, with the imagery degraded to coarser sensor resolutions by area averaging. This is the strongest available answer to *'will it work on our satellite data?'*

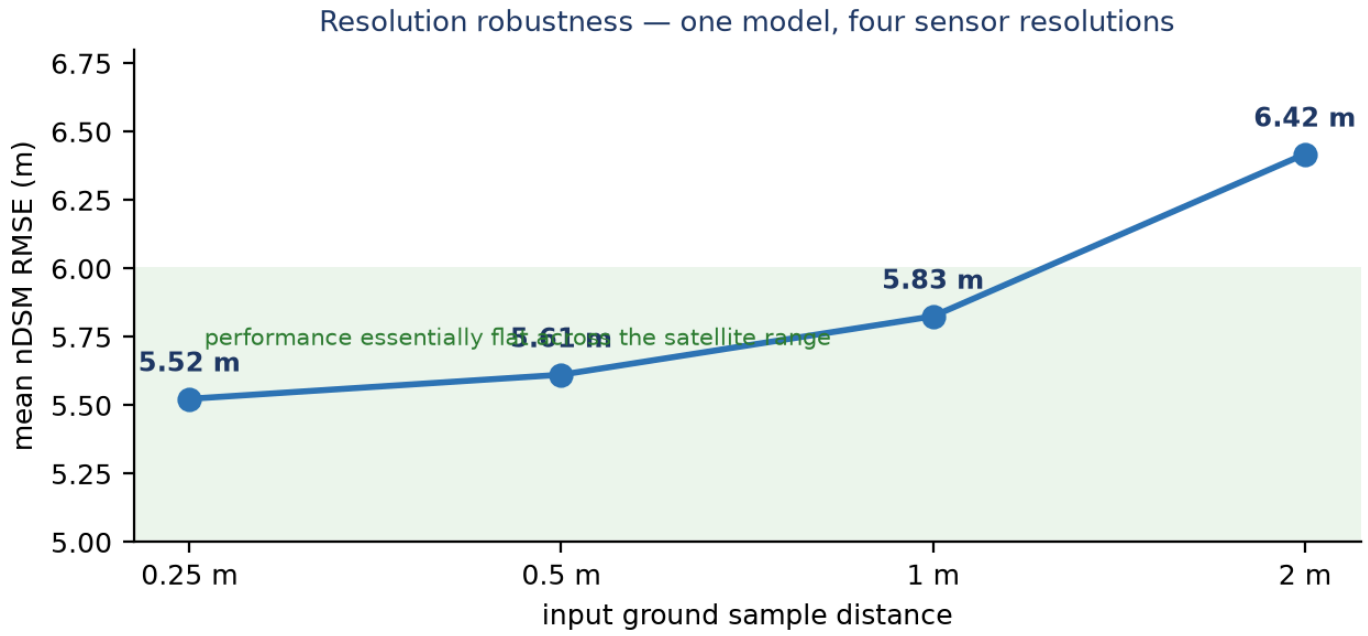


Figure 5 — Accuracy against input resolution. Flat from 0.25 m to 1 m; graceful, not catastrophic, at 2 m.

Input GSD	Mean RMSE	Change vs 0.25 m	Comparable sensor
0.25 m	**5.52 m**	+0.00 m	aerial / drone orthophoto
0.5 m	**5.61 m**	+0.09 m	WorldView-class panchromatic
1 m	**5.83 m**	+0.30 m	Cartosat-2 / high-res commercial
2 m	**6.42 m**	+0.89 m	Cartosat-1 / Sentinel-2 upsampled

Achieved by normalising every input to the 0.33 m training GSD before inference, so the model always sees features at the scale it was trained on. Upsampling is capped at 3× so we never invent detail that is not there.

Generated 29 September 2026 from the shipped checkpoint `decoder_best.pth` (Depth Anything V2 Base + multi-scale DPT head, epoch 28). Every accuracy figure in this document is read from `outputs/benchmark_ahn/results.json` in the repository and can be regenerated with `python scripts/benchmark_ahn.py`. Resolution figures from a dedicated sweep of the same model. Repository: github.com/Aryan241/Heimdall